

DR. OMAIA AL-OMARI

Assistant Professor • Applied & Responsible AI Researcher • Academic Leader

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PROFESSIONAL PROFILE

Assistant Professor in the Information Systems Department at Prince Sultan University with more than 18 years of academic and IT experience. Research and teaching connect applied artificial intelligence, trustworthy and explainable AI, information systems, and digital transformation. Experienced in curriculum development, quality assurance, conference leadership, editorial service, and translating complex research into practical, responsible systems.

APPLIED AI Machine Learning · NLP	TRUSTWORTHY AI XAI · Ethics · Governance	VISION & ANALYTICS Computer Vision · Healthcare	INFORMATION SYSTEMS EA · Digital Transformation
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ACADEMIC APPOINTMENTS

Assistant Professor, Information Systems — Prince Sultan University (PSU), CCIS <i>Riyadh, Saudi Arabia</i> - Teach Information Systems and digital transformation courses; supervise senior projects and cooperative training; advise students. - Contribute to curriculum, quality assurance, assessment, textbook review, examination coordination, and college-wide academic initiatives.	Aug 2024–Present
Lecturer / Faculty Member — Shaqra University, College of Computing and IT <i>Saudi Arabia</i> - Taught computing and information systems courses and supported academic advising, student activities, accreditation, and quality assurance. - Served in academic administration and contributed to program development and institutional service.	2014–2024
Graduate Research Assistant — Universiti Kebangsaan Malaysia (UKM) <i>Bangi, Malaysia</i>	2013–2014
Lecturer — University of Ha'il <i>Ha'il, Saudi Arabia</i>	2009–2012
<i>Earlier academic roles: Instructor, Jerash Private University (2009); Instructor, Ministry of Education, Jordan (2007–2009); Teaching Assistant, Yarmouk University (2006–2007).</i>	

EDUCATION

PhD in Computer Science (Artificial Intelligence) — Universiti Kebangsaan Malaysia (UKM) <i>Thesis: "Noun Verb Extraction and Hybrid Krill Herd as Feature Selection for Document Classification."</i>	2021
MSc in Information Technology and Computer Sciences — Yarmouk University <i>Specialization: Computer Information Systems. Thesis: "Evaluation of the Effect of Stemming on Clustering of Arabic Documents."</i>	2009
BSc in Information Technology and Computer Sciences — Yarmouk University <i>Specialization: Computer Information Systems.</i>	2006

SELECTED RESEARCH & PUBLICATIONS

Research themes: applied and explainable AI, computer vision, NLP and multimodal learning, healthcare analytics, cybersecurity, AI governance, and information systems. [Complete publication record on Google Scholar](#) ↗

- 2026** Y. Alshboul, O. Al-Omari, and A. Bsoul. "Phishing Detection Using CNN–RNN with Sparrow Search Optimization Algorithm." *Journal of Multiscale Modelling*.
- 2025** A. Omari, O. Al-Omari, T. Al-Omari, and S. M. Fati. "A Predictive Analytics Approach to Improve Telecom's Customer Retention." *Frontiers in Artificial Intelligence*, 8, 1600357. [DOI](#)
- 2025** D. Nimma et al. "Object Detection in Real-Time Video Surveillance Using Attention-Based Transformer-YOLOv8 Model." *Alexandria Engineering Journal*, 118, 482–495. [DOI](#)
- 2025** O. Al-Omari, O. Alkhatib, and T. Al-Omari. "CNN-Based Automated Detection of Metastatic Cancer in Histopathology Images." *Engineering, Technology & Applied Science Research*, 15(4), 24478–24485. [DOI](#)
- 2025** S. M. Fati and O. Al-Omari. "Deep Learning-Based Automated Segmentation of the Parcellated Corpus Callosum in Brain MRI." *Engineering, Technology & Applied Science Research*. [DOI](#)
- 2025** O. Al-Omari, A. Alyousef, S. Fati, F. Shannaq, and A. Omari. "Governance and Ethical Frameworks for AI Integration in Higher Education: Enhancing Personalized Learning and Legal Compliance." *Journal of Ecohumanism*, 4(2), 80–86.
- 2025** O. Al-Omari and T. Al-Omari. "Artificial Intelligence and Posthumanism: A Philosophical Inquiry into Consciousness, Ethics, and Human Identity." *Journal of Posthumanism*, 5(2), 458–469. [DOI](#)
- 2024** O. Al-Omari, A. Omari, E. Othman, and A. A. Alrashed. "MuSeRAN: Multimodal Sentiment Analysis with Recurrent Attention Networks." *International Conference on Decision Aid Sciences and Applications (DASA)*, 1–5.
- 2024** M. Allahham, A. A. Alrashed, O. Al-Omari, and W. I. Almajali. "Combination of Blockchain and Machine Learning for the Improvement of Decision Making and Secure Supply Chain Authentication Management." *DASA*, 1–4.
- 2023** N. A. Khan, O. M. Al-Omari, and S. M. Alshahrani. "An Empirical Study on the Future of Publication Repositories and Its Adaptability in Public Universities." *Computational Intelligence*, Springer, 823–829.
- 2020** N. A. Khan et al. "Internet of Things (IoT) Based Educational Data Mining (EDM) System." *Journal of Mechanics of Continua and Mathematical Sciences*, 15(3), 271–284.
- 2014** R. Mahafdah, N. Omar, and O. Al-Omari. "Arabic Part-of-Speech Tagging Using k-Nearest Neighbour and Naive Bayes Classifiers Combination." *Journal of Computer Science*, 10(9), 1865–1873.
- 2011** O. Al-Omari. "Evaluating the Effect of Stemming in Clustering of Arabic Documents." *Academic Research International*, 1(1), 284.

TEACHING & MENTORSHIP

Current and recent courses at Prince Sultan University include Introduction to Information Technology (IS101), Systems Analysis and Design (IS231), Enterprise Architecture (IS321), Quantitative Analysis (IS371), Change Management for Digital Transformation (IS452), Technology Trends in Digital Transformation (IS453), and Senior Project (IS499).

- Teaching approach: problem-based learning, case analysis, applied projects, and responsible use of emerging technologies.
- Mentorship: senior projects, academic advising, research development, and cooperative training supervision.
- Curriculum and quality: curriculum review, learning-outcome alignment, assessment coordination, and accreditation support.

ACADEMIC LEADERSHIP & PROFESSIONAL SERVICE

Track Chair — ICEGOV 2026 2026

International Conference on Theory and Practice of Electronic Governance

- Support the scholarly program and review process for an international digital-governance conference coordinated by UNU-EGOV.

Editorial Board Member — Babylonian Journal of Networking (BJN) 2026–Present

Peer review and academic publication quality

- Contribute to peer review and support the journal's academic quality and publication standards.

Chair — CCIS EXPO 2026, Prince Sultan University 2026

College-wide academic showcase

- **Prince Sultan University service:** CCIS Website Administrator; CCIS Curriculum Committee; Intercollegiate Committee; IS Curriculum Committee; Exam Coordination Committee 2026; IS Textbooks Update.
- **Shaqra University service:** Quality Assurance Committee; academic advising and student activities; accreditation and curriculum support.

COMMUNITY ENGAGEMENT & RECOGNITION

- **529 documented volunteer hours:** Saudi National Volunteer Work Platform, Ministry of Human Resources and Social Development; community activity recorded since 2021.
- **Practicing Quality Ambassador:** Quality Ambassadors Program, Membership No. 20230035, 2023.
- **Institutional recognition:** Recognized for contributions to Computer Science program accreditation at Shaqra University, 2023.
- **Graduate and undergraduate scholarships:** Scholarship awards from Jordan's Ministry of Higher Education and Yarmouk University.
- **Dean's Honor List:** Recognized across five semesters during undergraduate study.

SELECTED PROFESSIONAL DEVELOPMENT

Artificial Intelligence Analyst IBM Mastery Award	Artificial Intelligence Fundamentals IBM SkillsBuild
Course Design Institute Coursera · 2024	AWS Immersion Day Cloud Computing Workshop
Cisco Networking IoT & Cybersecurity	Agile Meets Design Thinking Coursera

LANGUAGES & AVAILABILITY

Languages: Arabic (native); English (professional proficiency).

Open to: research collaboration, academic leadership opportunities, and responsible AI initiatives.